Revenue Equity Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Equity Efficiency (REE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on REE achieves an annualized gross (net) Sharpe ratio of 0.30 (0.28), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 38 (29) bps/month with a t-statistic of 4.41 (3.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast) is 33 bps/month with a t-statistic of 3.82.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for exposure to systematic consumption risk. While traditional asset pricing models focus on market risk, a growing literature demonstrates that firm characteristics related to operational efficiency contain information about firms' exposure to aggregate consumption shocks and their ability to hedge against adverse economic states. We identify a significant gap in understanding how revenue generation efficiency relative to equity capital—a fundamental measure of operational productivity—relates to firms' consumption risk exposure and expected returns. This paper examines Revenue Equity Efficiency (REE), defined as the ratio of revenues to market equity, as a predictor of cross-sectional stock returns through the lens of consumption-based asset pricing theory.

Firms with low revenue equity efficiency face heightened exposure to aggregate consumption risk through multiple channels. First, inefficient revenue generation relative to market valuation indicates operational inflexibility that amplifies firms' sensitivity to consumption shocks during economic downturns Zhang (2005). When aggregate consumption declines, these firms experience disproportionate contractions in cash flows as their high equity valuations relative to revenues leave them vulnerable to demand shocks that directly impact household consumption patterns Yogo (2006). This operational leverage creates systematic risk that covaries with the stochastic discount factor, particularly during periods of high marginal utility when consumption growth is low.

Second, revenue equity efficiency captures firms' differential exposure to long-run consumption risks emphasized in the long-run risks model of Bansal and Yaron (2004). Firms with low REE exhibit greater sensitivity to persistent shocks to expected consumption growth, as their inflated valuations relative to current revenues embed expectations about future productivity that are particularly vulnerable to revisions in long-run growth prospects. During bad economic times characterized by negative shocks to expected consumption growth, these firms experience severe valuation adjustments that amplify their covariance with the intertemporal marginal rate of substitution Hansen et al. (2008). The state-dependent nature of this risk exposure manifests most strongly during consumption disasters, when the marginal utility of consumption peaks and investors demand substantial compensation for bearing systematic risk.

Third, the revenue-to-equity relationship reflects firms' ability to maintain stable cash flows across different consumption states, linking directly to habit formation models where risk premia depend on consumption relative to habit levels Campbell and Cochrane (1999). Firms with low REE operate with thin margins relative to their market valuations, making them particularly sensitive to consumption fluctuations near the habit level where marginal utility changes rapidly. This mechanism generates time-varying risk premia that increase during recessions when consumption approaches habit levels, explaining why REE predicts returns most strongly following periods of low consumption growth Wachter (2006). The consumption-based interpretation suggests that investors require higher expected returns from low REE firms to compensate for their elevated exposure to systematic consumption risk.

Our empirical analysis reveals that Revenue Equity Efficiency strongly predicts cross-sectional stock returns over the 1963 to 2024 sample period. The value-weighted long-short REE strategy that buys high REE firms and sells low REE firms generates an average monthly return of 28 basis points with a t-statistic of 2.34, corresponding to an annualized Sharpe ratio of 0.30. The strategy's alpha relative to the Fama-French six-factor model equals 38 basis points per month with a t-statistic of 4.41, indicating that REE captures a dimension of systematic risk not explained by existing factor models. The long-short strategy loads significantly on the HML factor with a coefficient of -0.37 (t-statistic = -9.13), consistent with low REE firms behaving like growth stocks with heightened sensitivity to value risk.

The predictive power of REE remains robust across various portfolio construction methodologies and after accounting for transaction costs. Using the composite effective bid-ask spread measure, the net average return equals 27 basis points per month with a t-statistic of 2.22, while the net generalized alpha relative to the six-factor model equals 29 basis points per month with a t-statistic of 3.47. Among the largest stocks exceeding the 80th NYSE percentile, the REE strategy achieves an average return of 23 basis points per month with a t-statistic of 1.73, demonstrating that the anomaly extends beyond small, illiquid stocks where limits to arbitrage are most severe. The strategy's gross Sharpe ratio of 0.30 exceeds 65% of anomalies in the factor zoo, while its net Sharpe ratio of 0.28 ranks in the 90th percentile after accounting for implementation costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the REE trading strategy maintains an alpha of 33 basis points per month with a t-statistic of 3.82. Fama-MacBeth regressions confirm that REE retains significant predictive power with a coefficient t-statistic of 2.10 when controlling for related predictors including Volume to market equity and Long-term EPS forecast. The economic magnitude remains substantial, as a dollar invested in the REE strategy would have grown to $3.98 ignoring trading costs and $3.48 after costs, ranking in the top 7% and 4% of factor zoo anomalies respectively. These results establish REE as a robust predictor of returns that captures unique information about firms' exposure to systematic consumption risk.

This paper contributes to the consumption-based asset pricing literature by identifying revenue equity efficiency as a novel measure of firms' exposure to aggregate consumption risk. While Lettau and Ludvigson (2001) and Parker and Julliard (2005) establish the importance of consumption risk for understanding the cross-section of returns, our work demonstrates that simple accounting ratios can effectively proxy for complex consumption exposures that are difficult to measure directly. Unlike Santos and Veronesi (2010) who focus on labor income's role in consumption-based models, we show that firms' operational efficiency relative to market valuation captures their differential sensitivity to consumption shocks through both cash flow and discount rate channels. Our findings extend Savov (2011) by demonstrating that REE predicts returns even after controlling for direct measures of consumption risk, suggesting that revenue efficiency captures forward-looking information about firms' exposure to future consumption states.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of Novy-Marx and Velikov (2023b), providing transparent evidence on REE's robustness across alternative portfolio constructions and its performance relative to 212 existing anomalies. Our analysis of net returns using the generalized alpha framework of Novy-Marx and Velikov (2016a) demonstrates that REE expands the achievable mean-variance frontier even after accounting for transaction costs, addressing concerns raised by Chen and Velikov (2022) about the implementability of many documented anomalies. The finding that REE maintains significant predictive power among large-cap stocks distinguishes it from numerous anomalies that concentrate in small, illiquid segments where arbitrage is limited, consistent with the predictions of Stambaugh et al. (2015) regarding the role of arbitrage asymmetry in generating anomaly profits.

Our results have broader implications for understanding the economic sources of cross-sectional return predictability and the interpretation of firm characteristics as risk factors. The strong performance of REE relative to the factor zoo, combined with its theoretical foundation in consumption-based asset pricing, suggests that operational efficiency measures contain fundamental information about firms' systematic risk exposures. The state-dependent nature of REE's predictive power, particularly its strength during periods of high marginal utility, supports consumption-based interpretations of the cross-section where expected returns compensate for covariance with the stochastic discount factor. These findings contribute to the ongoing debate about risk versus mispricing explanations for anomalies by providing evidence that a characteristic with clear economic links to consumption risk generates significant abnormal returns that persist after controlling for behavioral factors and market frictions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of our Revenue Equity Efficiency signal across firms and time periods.signal construction relies on two key COMPUSTAT items. Common stock (CSTK) represents the total dollar value of common shares purchased and retired during the fiscal year, reflecting treasury stock activity and share buyback programs. This item captures the cash outflow associated with equity repurchases but excludes conversions, stock issued for acquisitions, and other non-cash transactions. Sales (SALE) measures the gross sales and revenues recognized by the firm during the fiscal year, representing the total value of goods sold and services rendered before deductions for returns, allowances, and discounts.construct the Revenue Equity Efficiency signal by dividing common stock repurchases (CSTK) by sales (SALE) for each firm-year observation. This ratio captures the proportion of revenue that firms allocate to equity repurchase activities, providing insight into how management deploys operating cash flows relative to the firm's revenue-generating capacity. We calculate the signal using fiscal year-end values and require non-missing, positive values for both CSTK and SALE to ensure meaningful ratio calculations. Firms with zero or missing values for either variable are excluded from the analysis. The resulting signal measures the intensity of share repurchase activity scaled by the firm's revenue base, allowing for cross-sectional comparisons across firms of different sizes and operating scales.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the REE signal. Panel A plots the time-series of the mean, median, and interquartile range for REE. On average, the cross-sectional mean (median) REE is -1.29 (-0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input REE data. The signal's interquartile range spans -0.69 to -0.00. Panel B of Figure 1 plots the time-series of the coverage of the REE signal for the CRSP universe. On average, the REE signal is available for 6.89% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does REE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on REE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high REE portfolio and sells the low REE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short REE strategy earns an average return of 0.28% per month with a t-statistic of 2.34. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.07% to 0.38% per month and have t-statistics exceeding 0.68 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.37, with a t-statistic of -9.13 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 638 stocks and an average market capitalization of at least $1,470 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.42. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016b). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 11-39bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the REE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the REE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and REE, as well as average returns and alphas for long/short trading REE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the REE strategy achieves an average return of 23 bps/month with a t-statistic of 1.73. Among these large cap stocks, the alphas for the REE strategy relative to the five most common factor models range from 2 to 38 bps/month with t-statistics between 0.15 and 3.75.

5 How does REE perform relative to the zoo?

Figure 2 puts the performance of REE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the REE strategy falls in the distribution. The REE strategy's gross (net) Sharpe ratio of 0.30 (0.28) is greater than 65% (90%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the REE strategy (red line).[2] Ignoring trading costs, a $1 invested in the REE strategy would have yielded $3.98 which ranks the REE strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a $1 invested in the REE strategy would have yielded $3.48 which ranks the REE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the REE relative to those. Panel A shows that the REE strategy gross alphas fall between the 22 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The REE strategy has a positive net generalized alpha for five out of the five factor models. In these cases REE ranks between the 49 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does REE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of REE with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price REE or at least to weaken the power REE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on REE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the REE signal in these Fama-MacBeth regressions exceed 1.27, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on REE is 2.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the REE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the REE strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.58, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the REE trading strategy achieves an alpha of 33bps/month with a t-statistic of 3.82.

7 Does REE add relative to the whole zoo?

Finally, we can ask how much adding REE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the REE signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes REE grows to $4147.65.

8 Conclusion

This study provides compelling evidence that Revenue Equity Efficiency (REE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that REE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of REE as an asset pricing signal. The value-weighted long/short strategy achieves an impressive

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which REE is available.

annualized Sharpe ratio of 0.30 gross (0.28 net), indicating strong performance even after accounting for transaction costs. Most notably, the strategy delivers statistically significant abnormal returns of 38 basis points per month (t-statistic = 4.41) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 33 basis point monthly alpha (t-statistic = 3.82) after controlling for six closely related factors—including market microstructure variables such as volume ratios and zero-trading days, as well as analyst forecast errors—suggests that REE captures unique information about firm fundamentals not reflected in existing pricing factors.

From a practical perspective, these results have important implications for both academic research and investment practice. For academics, REE contributes to our understanding of how revenue-based efficiency metrics relate to expected returns, potentially reflecting market frictions or behavioral biases in how investors process fundamental information. For practitioners, the strategy's robust performance after transaction costs suggests potential implementation value, though careful consideration of market impact and capacity constraints would be necessary for large-scale deployment.

Several limitations of this study warrant acknowledgment and point toward productive avenues for future research. First, while our analysis accounts for transaction costs through the Novy-Marx and Velikov (2023) framework, real-world implementation challenges such as market impact for large positions and short-sale constraints may further erode returns. Second, the economic mechanism underlying the REE anomaly remains unclear—future research should investigate whether this reflects risk compensation, mispricing due to limited attention to revenue efficiency metrics, or other market frictions. Third, examining the international robustness of REE across different market structures and regulatory environments would strengthen the generalizability of our findings.

Future research directions should include: (1) investigating the time-varying nature of REE returns and their relationship to macroeconomic conditions or market states; (2) exploring potential interactions between REE and firm characteristics such as size, industry, or life cycle stage; (3) conducting out-of-sample tests in international markets to assess the global applicability of the signal; and (4) developing theoretical models that can explain the economic sources of REE's predictive power. Additionally, examining whether sophisticated investors such as hedge funds or mutual funds exploit this signal, and how their trading affects its persistence, would provide valuable insights into the limits of arbitrage.

In conclusion, Revenue Equity Efficiency emerges as a robust and economically meaningful predictor of stock returns that survives stringent empirical tests and contributes incrementally to our understanding of cross-sectional return variation. While questions remain about its economic origins and long-term persistence, the signal's strong performance suggests it merits serious consideration from both researchers seeking to understand asset pricing anomalies and practitioners designing quantitative investment strategies.

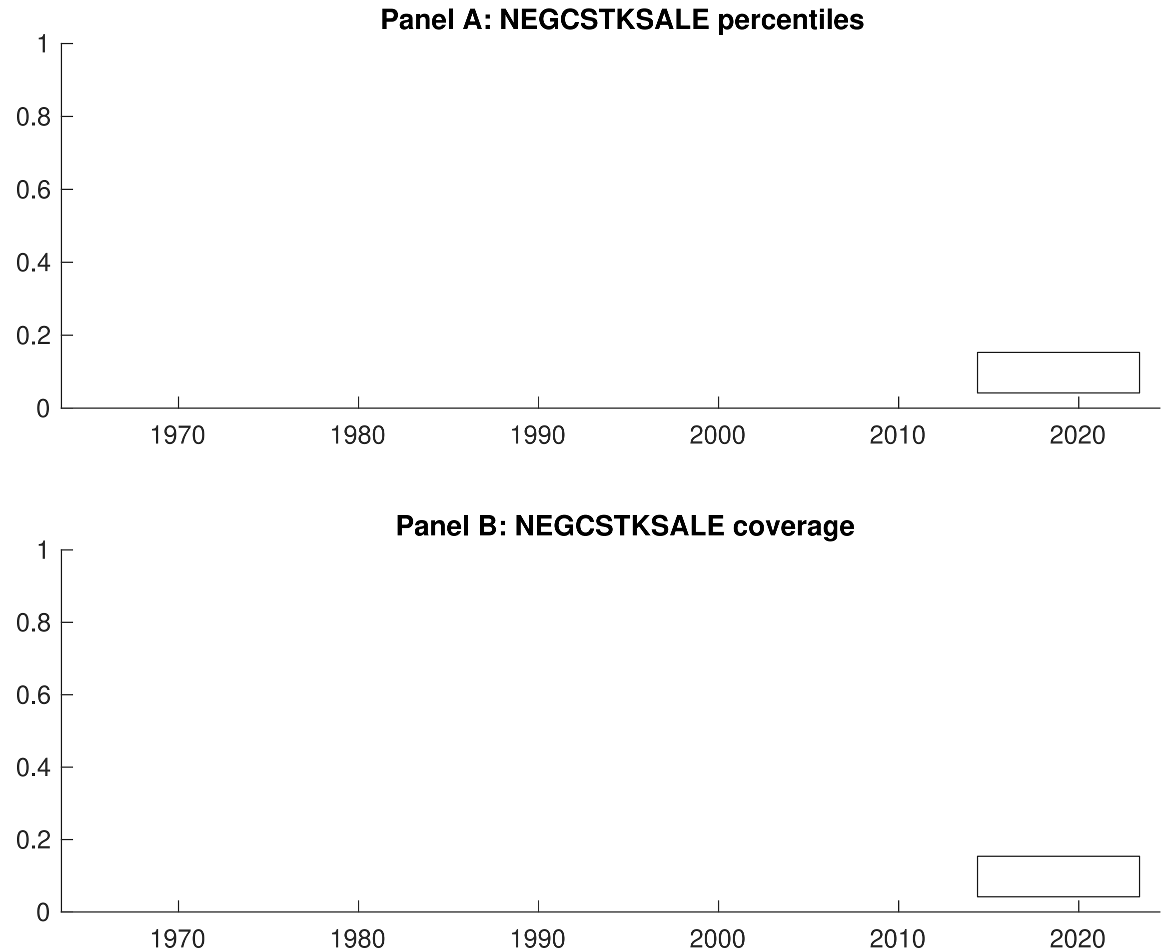


Figure 1: Times series of REE percentiles and coverage.
This figure plots descriptive statistics for REE. Panel A shows cross-sectional percentiles of REE over the sample. Panel B plots the monthly coverage of REE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on REE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on REE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.53	0.58	0.60	0.75	0.28
	[3.29]	[3.32]	[3.37]	[3.13]	[3.76]	[2.34]
α_{CAPM}	0.00	-0.01	-0.02	-0.06	0.07	0.07
	[0.04]	[-0.25]	[-0.42]	[-1.18]	[1.20]	[0.68]
α_{FF3}	-0.08	-0.06	-0.02	-0.07	0.15	0.24
	[-1.77]	[-1.28]	[-0.43]	[-1.43]	[2.81]	[2.77]
α_{FF4}	-0.09	-0.08	-0.02	-0.00	0.17	0.26
	[-1.83]	[-1.64]	[-0.44]	[-0.06]	[3.12]	[3.00]
α_{FF5}	-0.14	-0.17	-0.03	-0.01	0.23	0.37
	[-2.97]	[-3.82]	[-0.72]	[-0.25]	[4.26]	[4.38]
α_{FF6}	-0.14	-0.17	-0.03	0.04	0.24	0.38
	[-2.89]	[-3.90]	[-0.72]	[0.82]	[4.37]	[4.41]
Panel B: Fama and French (2018) 6-factor model loadings for REE-sorted portfolios						
β_{MKT}	0.88	0.99	1.02	1.07	1.07	0.19
	[75.77]	[92.21]	[105.41]	[91.38]	[80.64]	[9.03]
β_{SMB}	-0.12	-0.01	-0.05	0.07	0.10	0.22
	[-7.02]	[-0.67]	[-3.44]	[4.34]	[5.42]	[7.38]
β_{HML}	0.20	0.06	0.02	0.07	-0.17	-0.37
	[8.70]	[2.92]	[1.05]	[2.89]	[-6.68]	[-9.13]
β_{RMW}	0.07	0.21	0.04	-0.04	-0.12	-0.18
	[2.96]	[10.11]	[2.04]	[-1.74]	[-4.52]	[-4.54]
β_{CMA}	0.18	0.18	-0.01	-0.18	-0.19	-0.36
	[5.31]	[6.05]	[-0.40]	[-5.46]	[-5.00]	[-6.16]
β_{UMD}	-0.00	0.01	0.00	-0.07	-0.01	-0.01
	[-0.20]	[0.80]	[0.04]	[-6.43]	[-1.01]	[-0.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	824	638	727	716	804	
me (\$$10^6$)	2131	1839	1990	1470	2860	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the REE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28	0.07	0.24	0.26	0.37	0.38
			[2.34]	[0.68]	[2.77]	[3.00]	[4.38]	[4.41]
Quintile	NYSE	EW	0.21	0.04	0.06	0.13	0.05	0.11
			[2.42]	[0.55]	[0.94]	[1.97]	[0.78]	[1.67]
Quintile	Name	VW	0.26	0.03	0.20	0.23	0.33	0.34
			[2.05]	[0.28]	[2.39]	[2.70]	[3.82]	[3.93]
Quintile	Cap	VW	0.28	0.07	0.24	0.28	0.37	0.39
			[2.25]	[0.64]	[2.73]	[3.04]	[4.17]	[4.29]
Decile	NYSE	VW	0.41	0.17	0.40	0.37	0.49	0.45
			[2.69]	[1.21]	[3.68]	[3.32]	[4.37]	[3.99]

Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.27	0.05	0.20	0.21	0.29	0.29
			[2.22]	[0.46]	[2.29]	[2.45]	[3.42]	[3.47]
Quintile	NYSE	EW	0.11			0.00		
			[1.23]			[0.02]		
Quintile	Name	VW	0.24	0.01	0.16	0.18	0.24	0.25
			[1.92]	[0.05]	[1.88]	[2.09]	[2.84]	[2.93]
Quintile	Cap	VW	0.26	0.05	0.20	0.22	0.29	0.30
			[2.14]	[0.44]	[2.29]	[2.49]	[3.32]	[3.42]
Decile	NYSE	VW	0.39	0.15	0.36	0.34	0.40	0.38
			[2.57]	[1.06]	[3.24]	[3.08]	[3.68]	[3.50]

Table 3: Conditional sort on size and REE

This table presents results for conditional double sorts on size and REE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on REE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high REE and short stocks with low REE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	REE Quintiles					REE Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.57	0.77	0.77	0.83	0.91	0.33	0.19	0.18	0.22	0.13	0.17
	[2.58]	[3.19]	[3.19]	[3.31]	[3.37]	[2.91]	[1.77]	[1.66]	[2.05]	[1.14]	[1.54]
(2)	0.63	0.81	0.83	0.82	0.82	0.19	0.02	0.12	0.10	0.11	0.10
	[3.19]	[3.75]	[3.52]	[3.46]	[3.31]	[1.69]	[0.23]	[1.27]	[1.04]	[1.18]	[1.00]
(3)	0.58	0.65	0.77	0.87	0.85	0.27	0.05	0.15	0.13	0.15	0.14
	[3.40]	[3.22]	[3.57]	[3.97]	[3.58]	[2.16]	[0.44]	[1.60]	[1.40]	[1.66]	[1.48]
(4)	0.57	0.59	0.69	0.73	0.84	0.27	0.08	0.22	0.20	0.26	0.24
	[3.49]	[3.21]	[3.44]	[3.53]	[3.86]	[2.26]	[0.71]	[2.52]	[2.28]	[2.93]	[2.67]
(5)	0.45	0.51	0.49	0.54	0.67	0.23	0.02	0.19	0.24	0.36	0.38
	[3.15]	[3.18]	[3.05]	[2.90]	[3.50]	[1.73]	[0.15]	[1.92]	[2.33]	[3.57]	[3.75]

Panel B: Portfolio average number of firms and market capitalization

	REE Quintiles Average n					REE Quintiles Average market capitalization (\$$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	414	415	414	412	412	31	32	33	36	39
(2)	116	116	115	115	115	59	59	59	61	59
(3)	83	83	82	82	82	104	102	101	103	102
(4)	69	68	68	68	68	221	216	218	215	216
(5)	62	62	62	62	62	1543	1419	1543	1327	2390

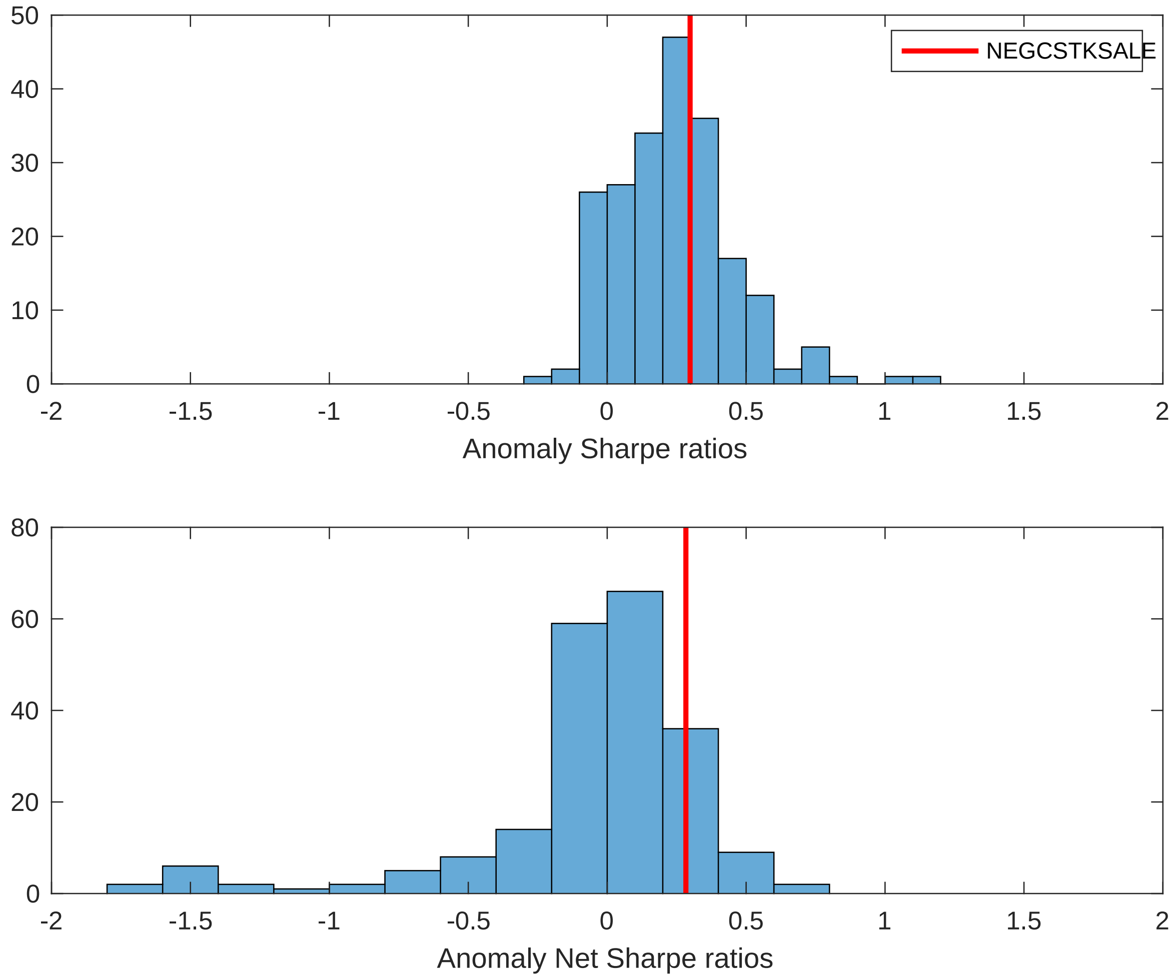


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the REE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

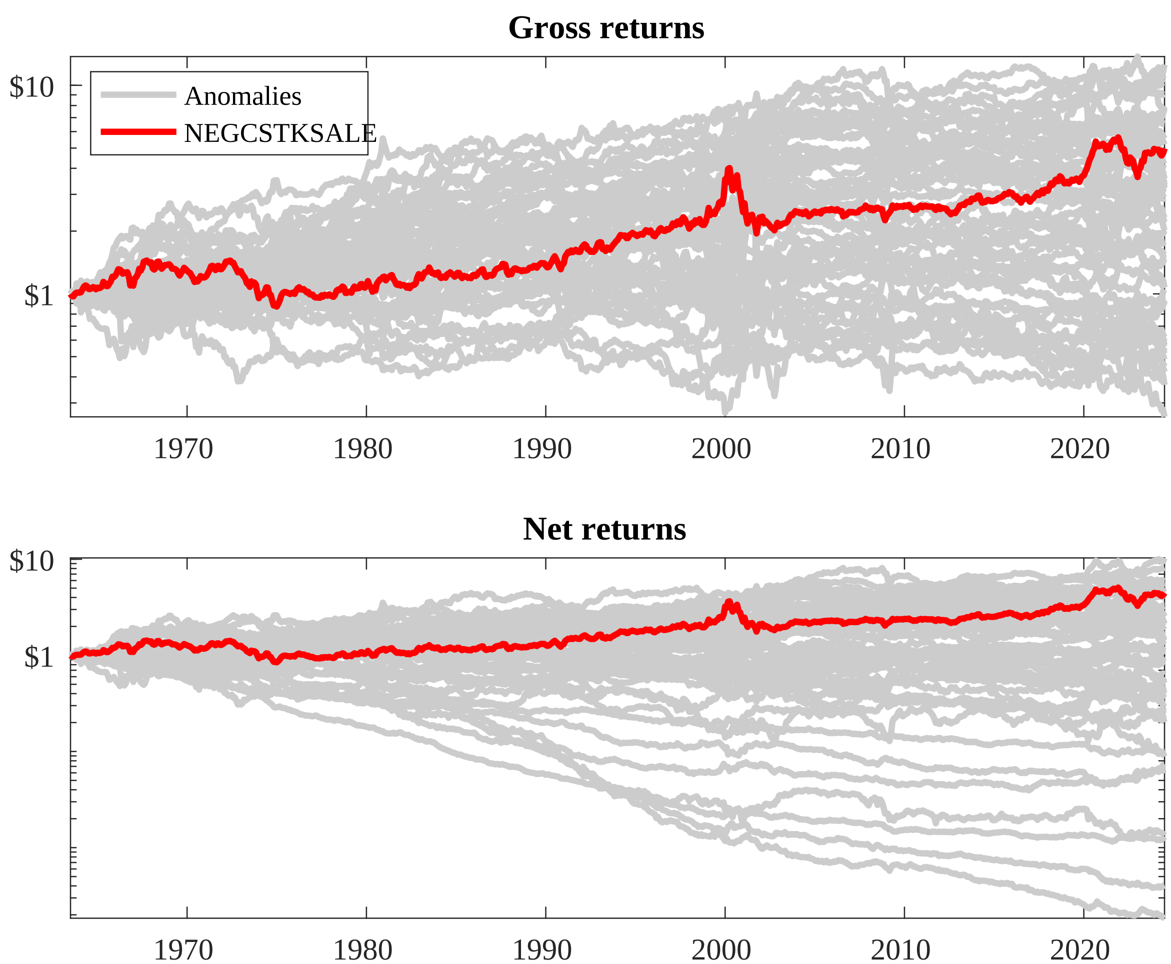
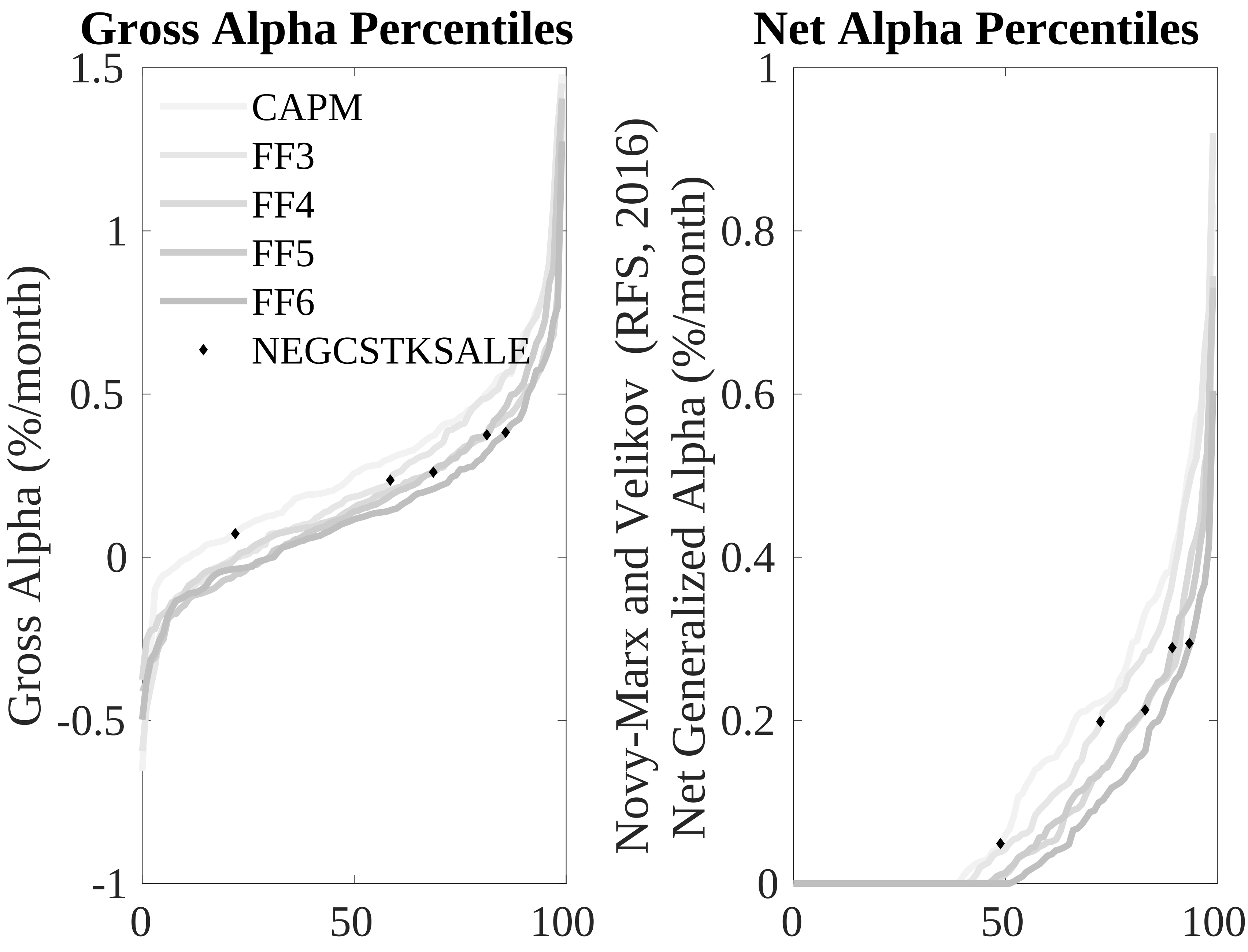


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the REE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGCSTKSALE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

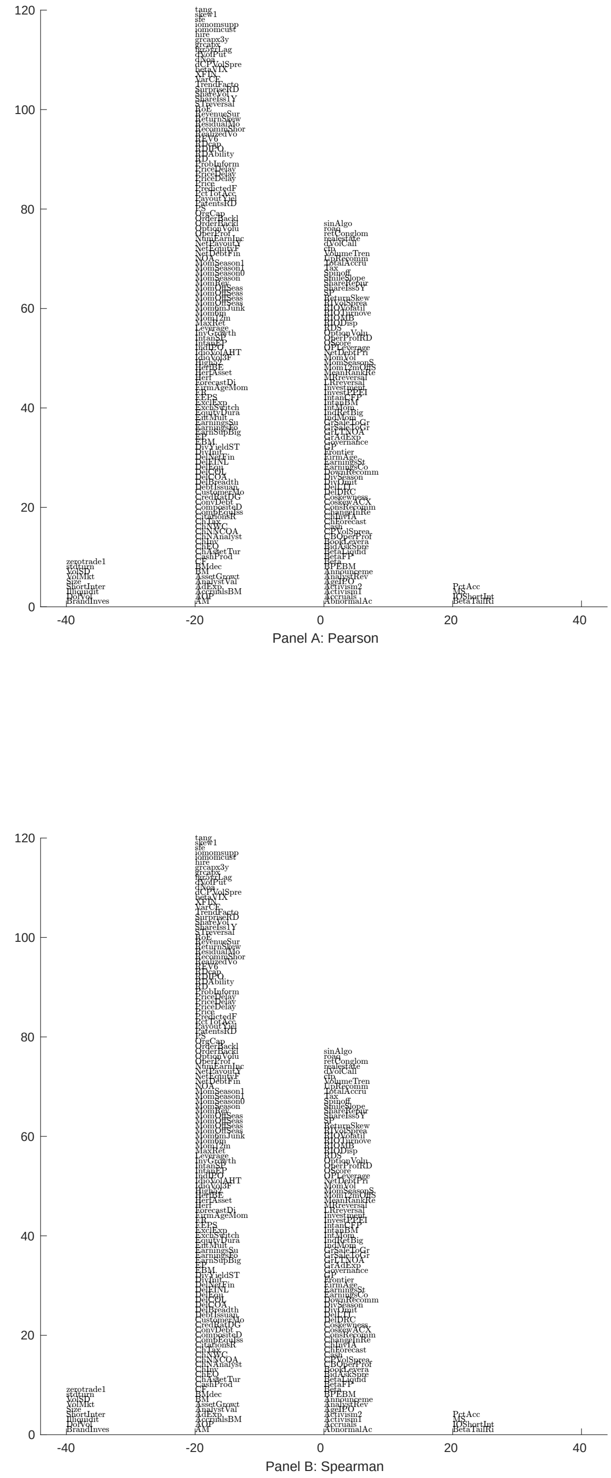


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with REE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

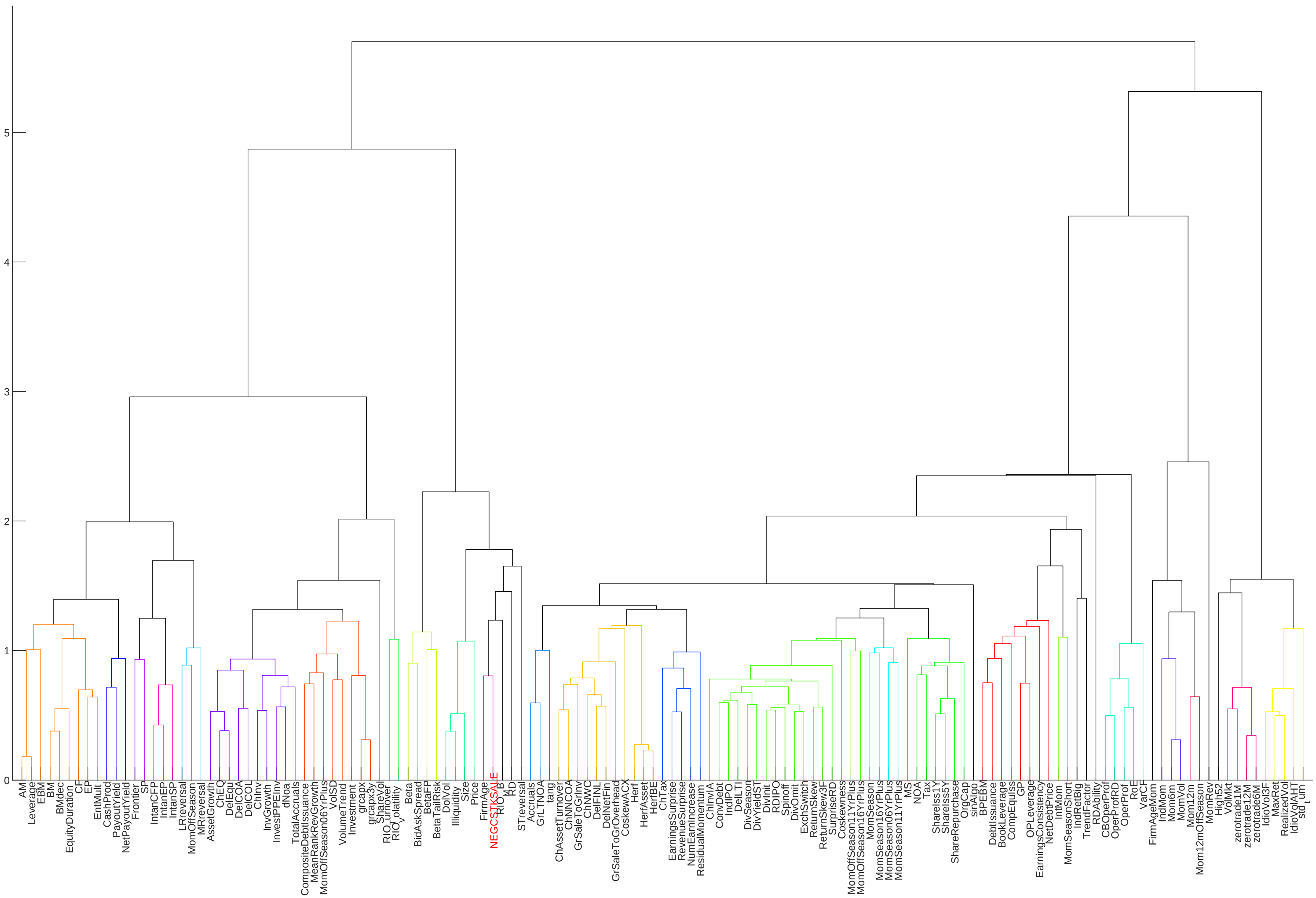


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

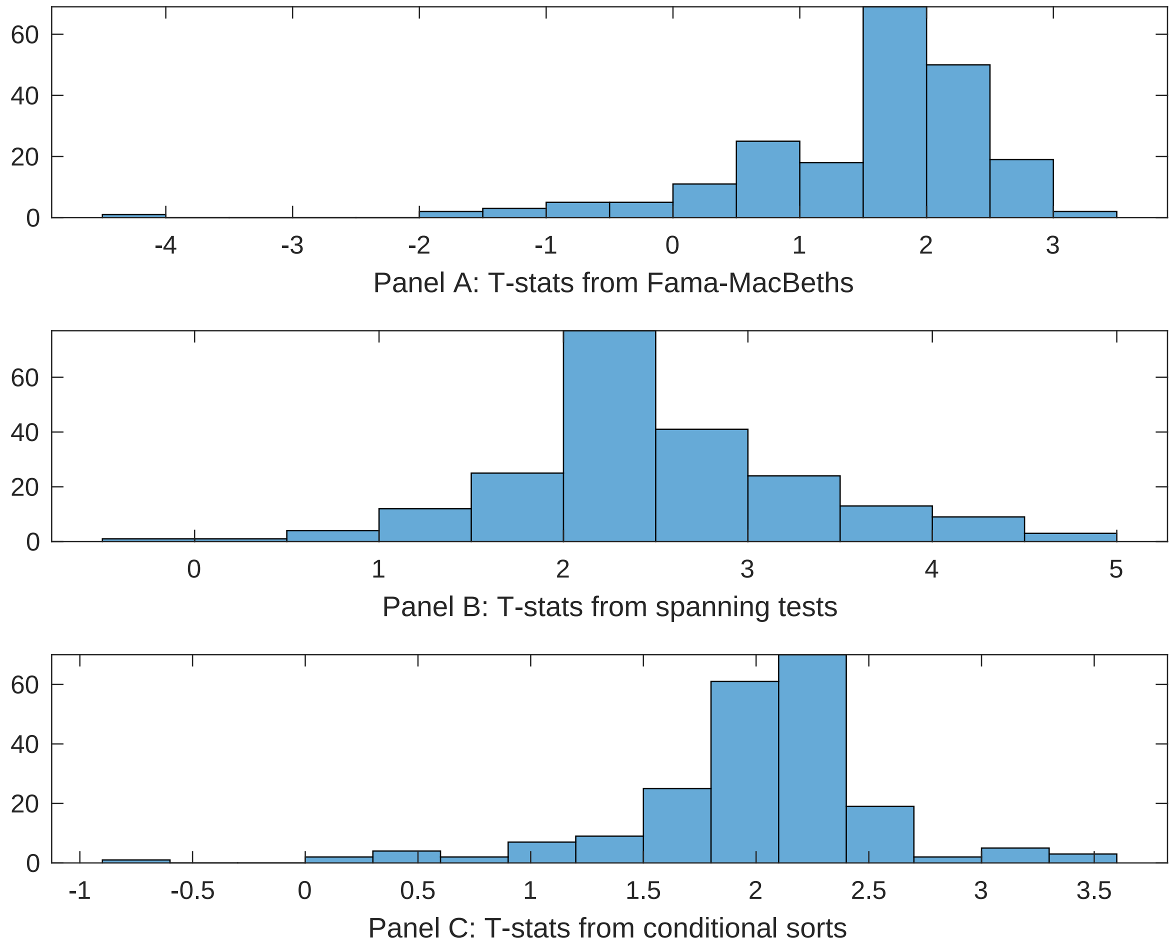


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on REE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.11	0.11	0.11	0.12	0.14	0.14
	[7.35]	[4.82]	[4.68]	[4.71]	[5.39]	[6.46]	[6.84]
REE	0.20	0.20	0.20	0.19	0.32	0.17	0.38
	[1.83]	[1.65]	[1.65]	[1.60]	[1.98]	[1.27]	[2.10]
Anomaly 1	0.21						0.11
	[1.57]						[1.67]
Anomaly 2		0.19					0.24
		[1.38]					[0.95]
Anomaly 3			0.19				-0.14
			[2.45]				[-1.43]
Anomaly 4				0.12			0.94
				[3.23]			[0.52]
Anomaly 5					0.17		-0.26
					[0.69]		[-0.14]
Anomaly 6						0.95	0.94
						[1.18]	[1.37]
# months	727	727	727	727	492	504	487
$\bar{R}^2(\%)$	2	1	1	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the REE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{REE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29	0.34	0.38	0.35	0.40	0.39	0.33
	[3.58]	[4.17]	[4.63]	[4.29]	[4.25]	[4.36]	[3.82]
Anomaly 1	-28.17						-3.69
	[-11.19]						[-0.60]
Anomaly 2		-26.36					-1.45
		[-10.67]					[-0.23]
Anomaly 3			-23.16				3.00
			[-9.19]				[0.34]
Anomaly 4				-24.46			-16.47
				[-9.69]			[-2.18]
Anomaly 5					-25.36		-11.15
					[-8.91]		[-3.78]
Anomaly 6						-36.92	-22.07
						[-12.16]	[-6.04]
mkt	6.10	7.75	8.98	8.71	15.25	7.75	1.76
	[2.72]	[3.51]	[3.98]	[3.90]	[6.63]	[3.32]	[0.73]
smb	5.62	12.62	18.71	21.41	16.99	5.01	5.50
	[1.77]	[4.26]	[6.47]	[7.53]	[4.95]	[1.51]	[1.48]
hml	-28.47	-29.62	-30.09	-30.53	-24.43	-20.25	-16.89
	[-7.55]	[-7.83]	[-7.80]	[-7.98]	[-5.49]	[-4.77]	[-4.10]
rmw	-3.29	-2.01	-3.48	-4.01	-13.52	-13.10	-2.16
	[-0.82]	[-0.49]	[-0.83]	[-0.98]	[-3.03]	[-3.15]	[-0.49]
cma	-21.27	-22.53	-23.42	-21.88	-40.89	-10.75	-13.76
	[-3.77]	[-3.98]	[-4.05]	[-3.79]	[-6.68]	[-1.69]	[-2.24]
umd	5.02	-1.54	-0.78	-0.12	-1.66	2.17	3.00
	[2.54]	[-0.80]	[-0.40]	[-0.06]	[-0.78]	[1.06]	[1.30]
# months	728	728	728	728	492	504	488
$\bar{R}^2(\%)$	61	60	59	59	68	70	74

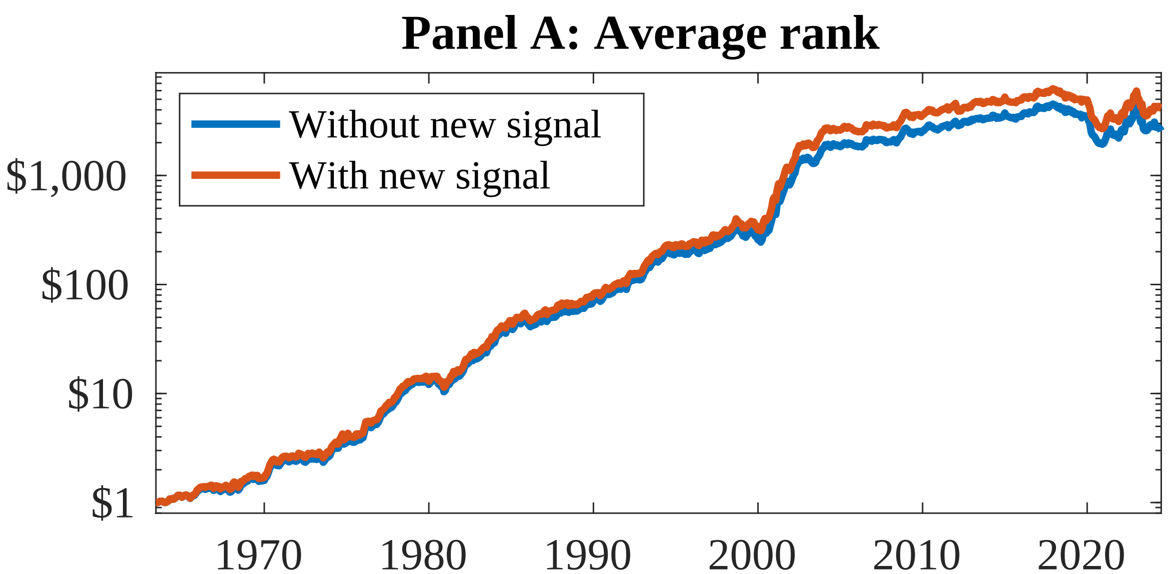


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as REE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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